

*Teaching Notes 3A**

Beta Management

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*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

Objectives

- Introductory case on risk and diversification
- To gain practice in calculating risk and return measures on stocks and portfolios; to calculate measures of beta for stocks.
- To develop an understanding of total risk, portfolio risk, diversifiable and non-diversifiable risk
 - Relate these to beta.
 - To “slowly” establish a relationship between risk and return that leads us towards developing the CAPM equation (Lecture 4A)

Investment Strategy of Beta Management

- Beta is a small (\$25M) fund management operation:
 - *“Beta Management’s stated goals were to enhance returns but reduce risks for clients via market timing” [p.1]*
- Invests in money market securities (risk-free) and Vanguard’s Index 500 Trust.
 - Currently 79.2% invested in the Vanguard Fund [p.1]
- Ms. Wolfe is thinking of switching \$200,000 from money markets to individual equities.
 - Decision is whether to invest in California REIT or Brown.
 - Share prices had come down and “looked cheap”
 - But prices are **volatile**, and Brown is sensitive to the market.
- How should we think about **“market timing”**?

Adding Value By Being a “Market Timer”?

Are Markets Efficient?

“A capital market is said to be efficient if it fully reflects all information in determining security prices. **Formally, the market is said to be efficient with respect to some information set...if security prices would be unaffected by revealing that information to all participants.** Moreover, efficiency with respect to an information set...implies that it is impossible to make economic (abnormal) profits by trading on the basis of that information set.”

Burton Malkiel, 1992

“A Random Walk Down Wall Street”

Is Ms. Wolfe A Technical Analyst?

- **Technical analysis** refers to the attempt to predict stock price movements by examining patterns in past stock prices.

“The technical analyst may go even further...He may suggest that he could trade with profit in a stock knowing only its ticker symbol, completely ignorant of the company, the industry, what it manufactures or sells...if your market technician is really experienced at his business, he could, in theory, do exactly what he claims.”

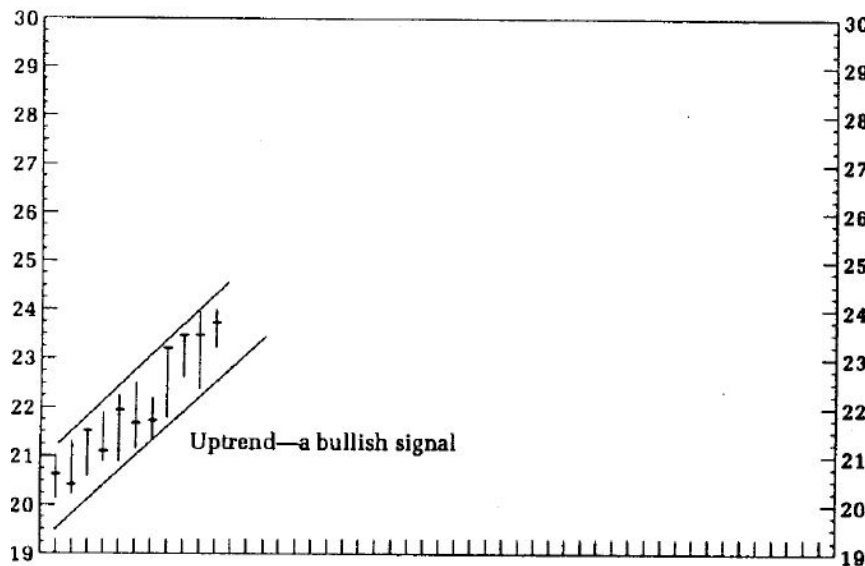
R.D. Edwards

“Technical Analyst of Stock Trends”

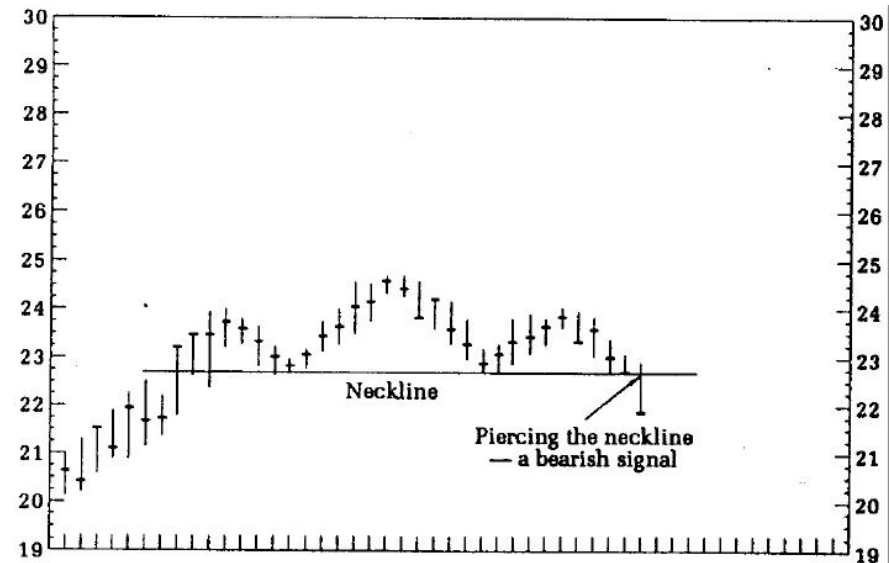
Is Ms. Wolfe A Technical Analyst? (cont.)

- Easy to look for patterns, so regularities are quickly exploited.
 - Some patterns in stock prices that technical analysts look for are: Buy Low, Sell High, Trends/Momentum, and Head and Shoulder.
 - Why are all of us not technical analysts then?

Trends – Momentum



Head And Shoulder



Starting from “where we left off in Lecture 2B”

Total Risk: Market + Firm Specific Risk

- The **total risk** of a security is made up of the risk from market movements (market risk) and the risk from firm-specific movements (firm-specific risk).
- Two securities can have the **same** total risk, yet that risk can be made up of very **different** components.
 - Standard deviation is a measure of the **total risk** of each stock.

Monthly	Vanguard Index	California REIT	Brown Group
Mean	1.103%	-2.265%	-0.671%
Std Dev	4.606%	9.231%	8.167%
Covariance		0.000	0.002
Correlation		0.074	0.656

- REIT and Brown are riskier in total risk sense than the Vanguard Index Trust 500, which is our “market proxy”.

Monthly vs. Annual Standard Deviations

- Annualized Standard Deviation is usually called **volatility**:
 - Multiply the monthly volatility by the square-root of 12.

Annual	Vanguard Index	California REIT	Brown Group
Mean	13.23%	-27.19%	-8.06%
Std Dev	16.0%	32.0%	28.3%

- The two stocks have volatilities about **twice** the market.
- **Bottom Line:** By switching from the index fund to the individual stocks we are switching into higher risk assets.
 - This would seem to imply that the risk borne by Beta Management's clients will **increase....???**

Expected Returns

- Based on this measure of risk, we would expect that California REIT is riskier than Brown Group by 14% (32.0%/28.3%).
 - This would suggest that it should have a higher **expected return**.
- Both stocks are riskier than the market.
 - Historically California REIT and Brown Group have been bad investments.
 - Over the last 2 years, market has been up by 13.2%!
 - California REIT is down 27.2% per year: hit by California earthquake and weak California real estate market.
 - Brown is down 8.1% per year: Earnings fell due to recession.

Portfolio Risk

- What matters to the investors is **not** whether a risky stock is added to the portfolio.
 - What matters is whether the resulting portfolio is more risky.
- The final risk of the portfolio depends on how stock(s) which are added “get along” with the original stocks in the portfolio.
- There are some stocks, which are risky on their own, but which are negatively correlated with the rest of the portfolio and hence reduce the portfolio risk when added.
 - **Diversification** can reduce risk even if stocks not **negatively** correlated...is enough that not **perfectly** positively correlated!
 - Do not examine assets in **isolation! Covariances matter!!**

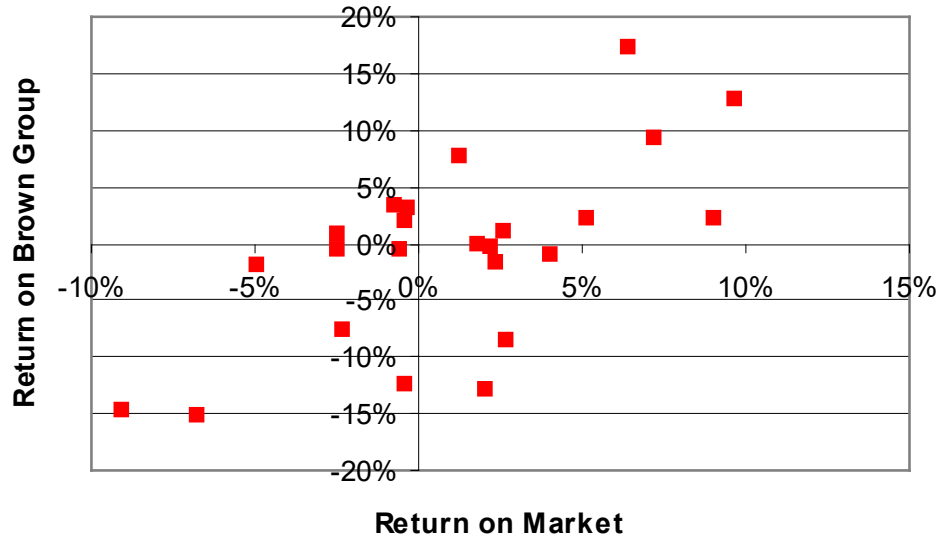
Portfolio Risk: Shifts In Asset Allocation

- Two Shifts:
 - 1% of their portfolio to Brown
 - 1% of their portfolio to California REIT.

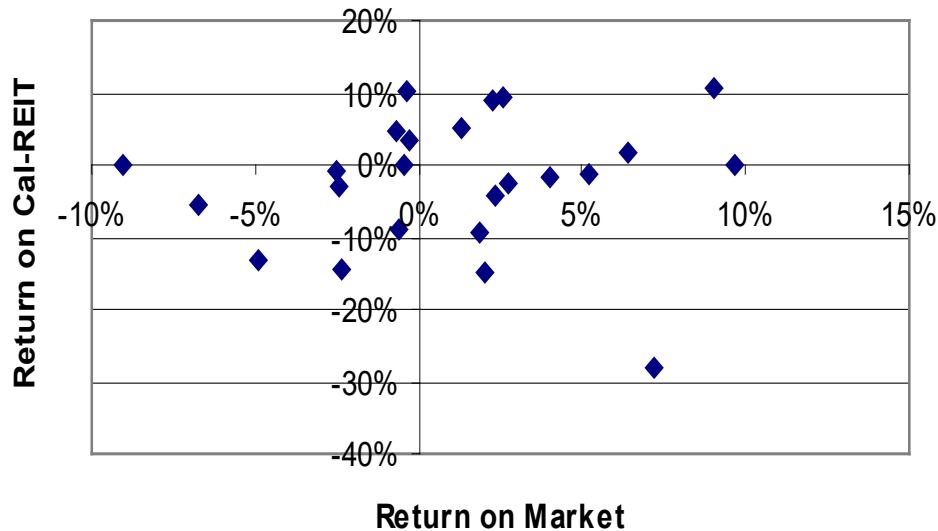
Monthly	100% Index	99% Index 1% REIT	99% Index 1% Brown
Mean	1.103%	1.069%	1.085%
Std Dev	4.606%	4.568%	4.614%

- California REIT, while individually riskier than Brown, actually reduces risk when added to the portfolio because its correlation with the portfolio (Index) is **very low**.
 - It has **less** market risk! It has **higher** diversifiable risk!
- Brown is **highly** correlated with the Index.
 - It has **higher** non-diversifiable or market risk!

Which Investment Has Higher Market Risk?



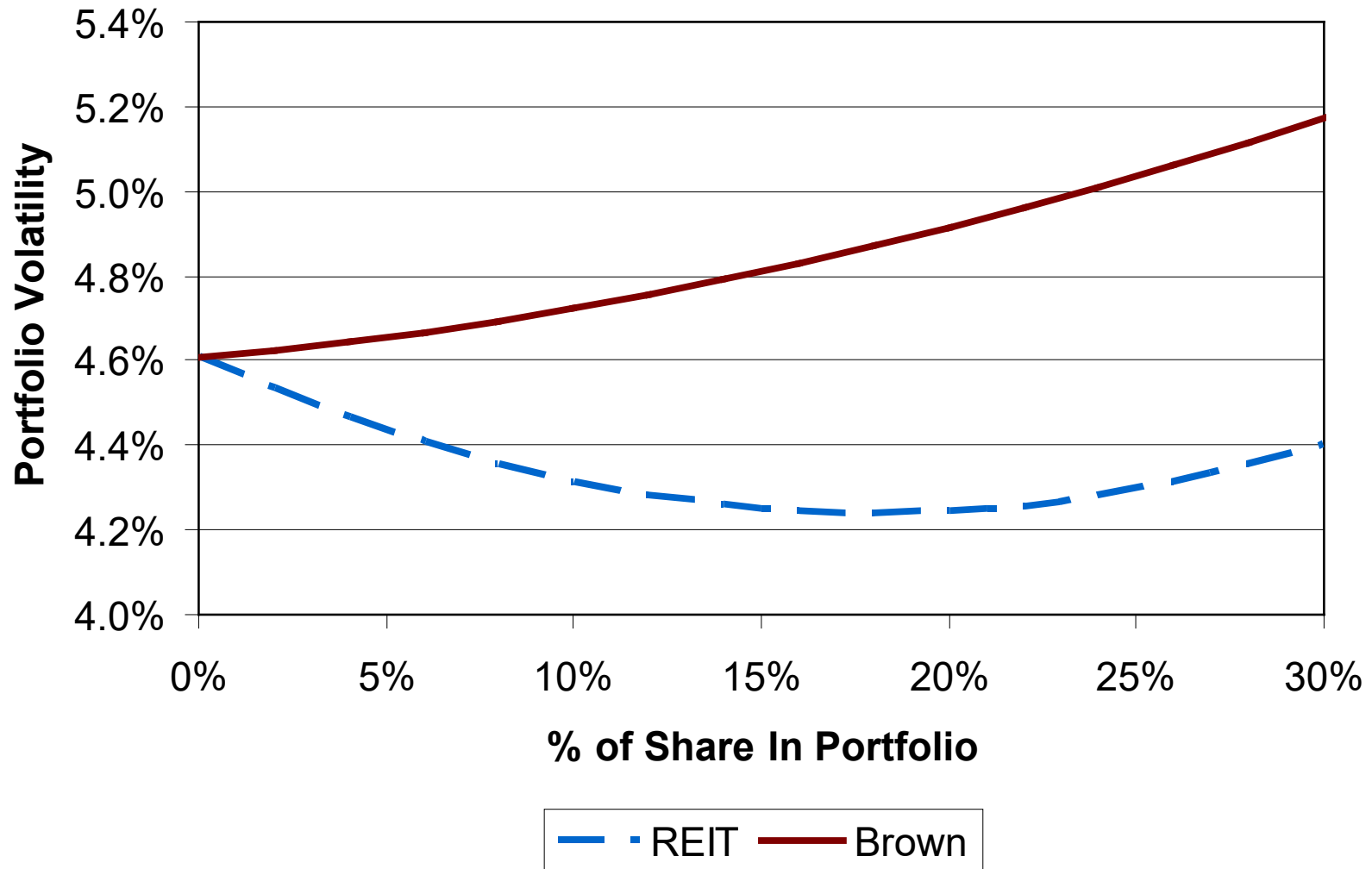
Correlation = 0.66



Correlation = 0.07

Portfolio Risk: Different Asset Allocations

Portfolios With Different Asset Allocations



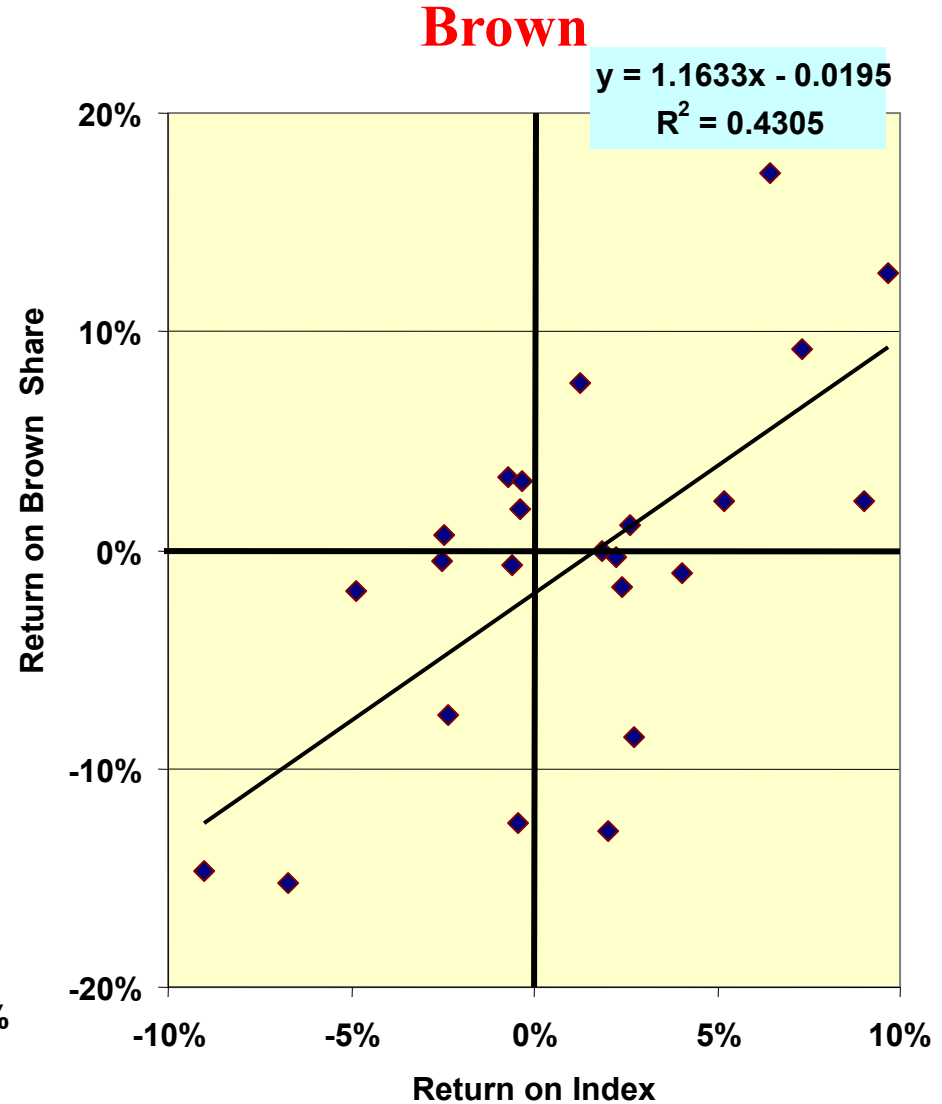
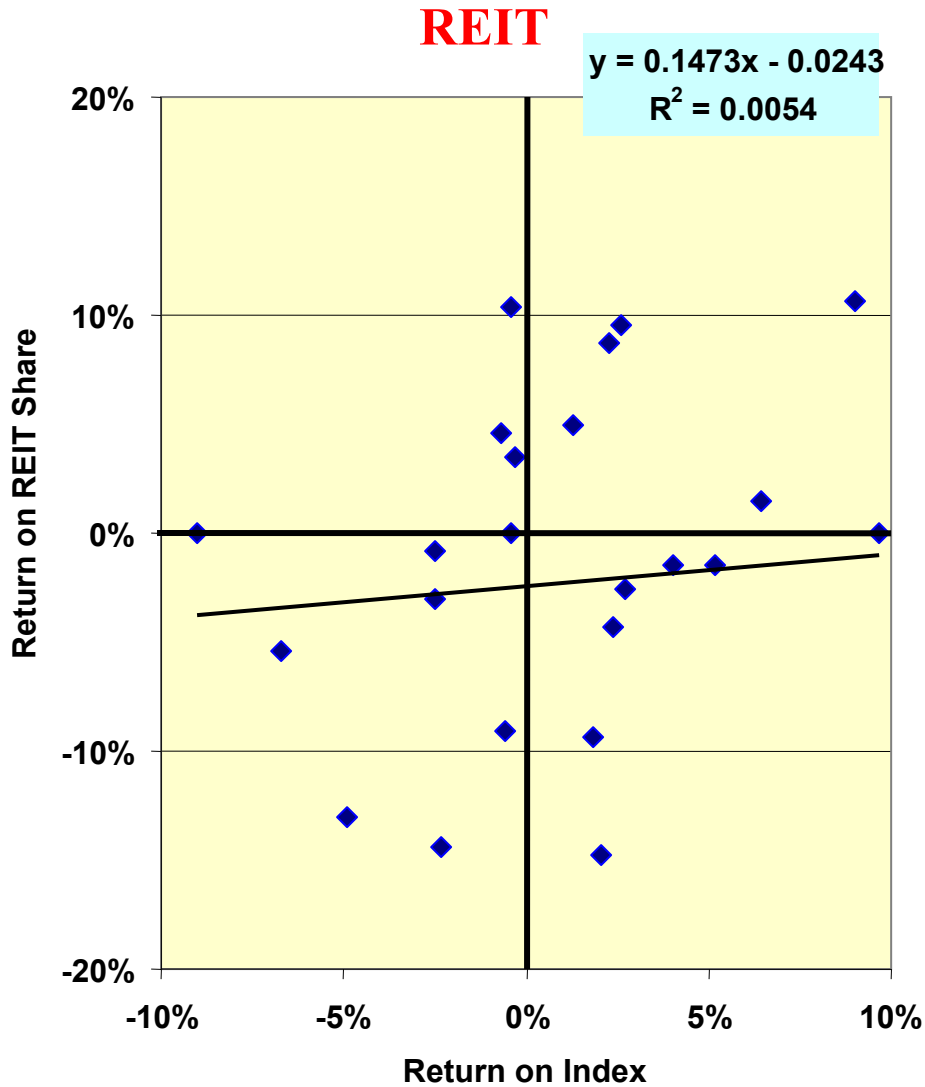
Market Risk Is Measured By Beta

- If you want to know the contribution of an individual security to the risk of a well-diversified portfolio:
 - Do **not** think about how risky that security is if held in isolation.
 - You need to measure its **market risk**: how sensitive is that stock to market movements.
- Beta is a measure of how an asset will contribute to the risk of a well-diversified portfolio.
 - Beta measures its **sensitivity** to market movements.
- Stocks with:
 - $\beta > 1$ tend to **amplify** the overall movements of the market.
 - $0 < \beta < 1$ move in the **same** direction as the market (not as much)
 - $\beta < 0$ move in **opposite** direction to the overall market.

Volatility vs. Beta

- **If had measured Risk using volatility...**
 - Both investments have similar risks, higher than the market.
 - Therefore, given that both have similar total risk, you would attribute high risk premium to both shares.
- The composition of the total risk is different for both stocks:
 - California REIT has **less** market risk and **more** diversifiable risk.
 - Brown Group has **more** market risk and **less** diversifiable risk.
- **Should be measuring Risk using Beta!!**
 - It takes into account only market risk.
 - Most risk of California REIT is diversifiable → **Low Beta!**
 - Some risk of Brown is diversifiable; it has a beta just over one even though its volatility is almost double that of the market.

Volatility versus Betas: Regressions

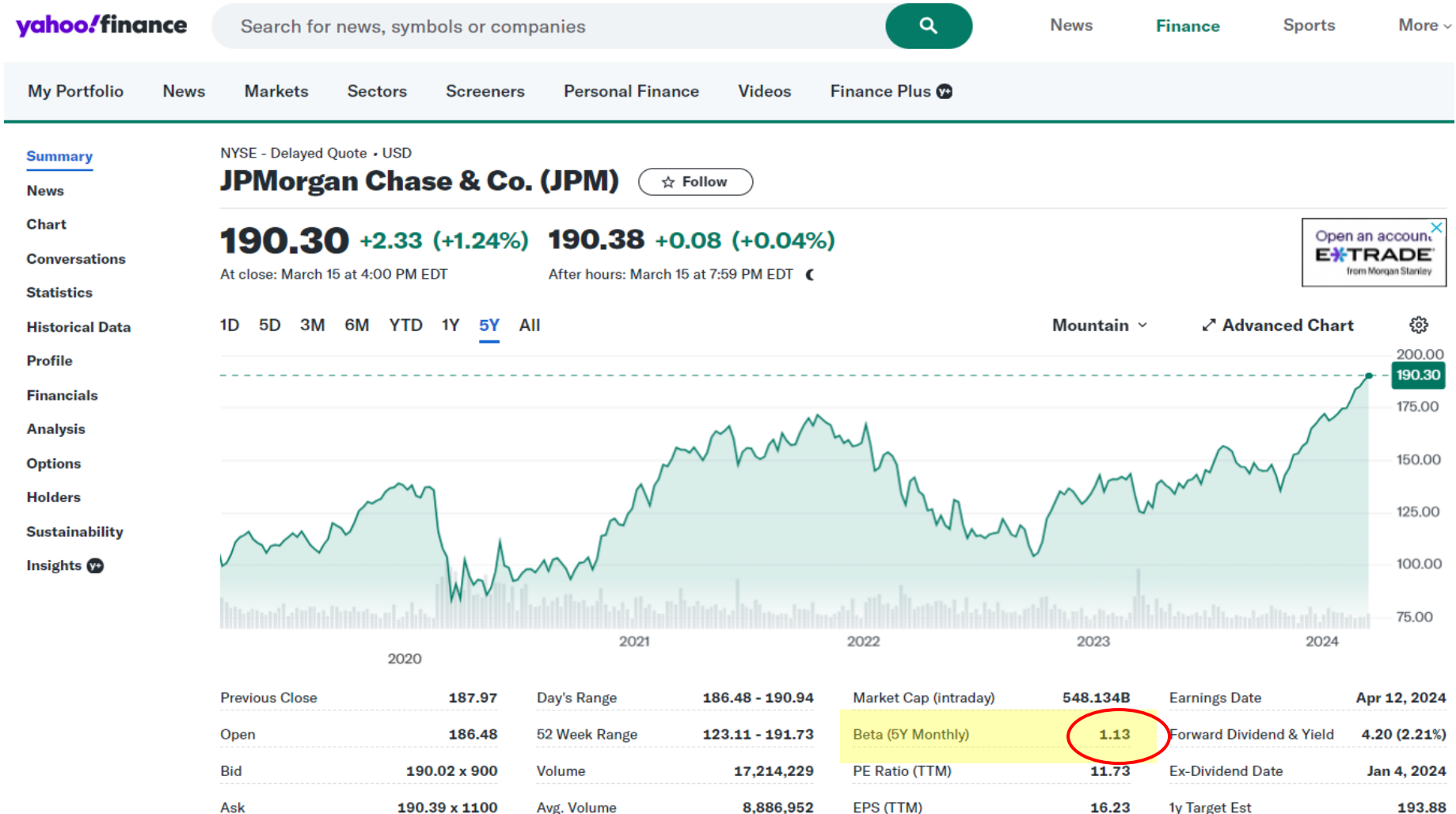


Volatility vs. Beta: Bottom Line

- The market has a $\beta = 1$
 - If we add an asset (Brown Group) with $\beta > 1$, the risk of the portfolio **increases** from 4.606% to 4.614%!
 - If we add an asset (California REIT) with $\beta < 1$, the risk of the portfolio **decreases** from 4.606% to 4.568%!
 - Even if the asset added is riskier than our portfolio (in terms of total risk).
- The Betas of the new portfolios are given by:
$$\beta_{\text{Portfolio}} = \alpha_1 \times \beta_{\text{Market}} + \alpha_2 \times \beta_{\text{Stock}}$$
$$\beta_{\text{Portfolio w/ Brown}} = 0.99 \times 1.00 + 0.01 \times 1.16 = 1.0016$$
$$\beta_{\text{Portfolio w/ REIT}} = 0.99 \times 1.00 + 0.01 \times 0.15 = 0.9915$$
- **Bottom Line:** The required return is related to **non-diversifiable risk**, measured by beta!

Beta as Measure of a Stock Risk

- Even Yahoo Finance reports a stock's beta, together with price, market cap, P/E ratio etc. (but not its volatility)



Why This Case?

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 - The final risk of the portfolio depends on how the stock(s) which are added “get along” with the original stocks in the portfolio.
- There are some stocks, which are risky on their own, but which are negatively correlated with the rest of the portfolio and hence reduce the portfolio risk when added.
 - **Diversification** can reduce risk even if stocks are **not negatively** correlated.
 - A comprehensive measure of risk that accounts for this is **beta**.

Why This Case? (cont.)

- Things to remember about **beta**:
 - Measures the extra risk that the shares contribute when they are added to a diversified portfolio.
 - Gives a different signal about risk than volatility.
 - Depends on the correlation of the share with the market portfolio.
 - Ignores risk that is diversifiable.
 - It is measured by the slope of the regression of the share's returns on the market returns.